Utility of iPhone-Based Pupillometry in Comparing Pupillary Dynamics Between Sport Concussed Subjects With Photosensitivity and Healthy Controls

Pritam Dutta, MOptom, FAAO

Background: To compare the pupillary dynamics using an iPhone-based pupillometry technique in subjects with sports concussion with photosensitivity and aged-matched controls.

Methods: Fifty subjects with sports concussion were compared with 50 aged-matched healthy controls. Athletes with persistent concussive symptoms for 1 year or more after the initial injury were included. All the subjects underwent a Post-Concussion Symptom Scale (PCSS) administration followed by pupillary dynamics measurement using an iPhone-based application (Reflex-Pro PLR analyzer).

Results: The mean age was 27 ± 4 years in the concussed group and 26 ± 5 years in the control group. In subjects with concussion, there was a significant decrease in the mean of the following parameters: average constriction speed (1.10 ± 0.15 vs 1.78 ± 0.12 mm/s; $P < 0.001$), maximum constriction speed (2.05 ± 0.26 vs 3.84 ± 0.28 mm/s; $P < 0.001$), average diameter (3.64 ± 0.12 vs 0.36 ± 0.05 mm; $P < 0.001$), maximum diameter (4.75 ± 0.17 vs 5.23 ± 0.16 mm; $P < 0.001$), and minimum diameter (2.75 ± 0.17 vs 3.64 ± 0.11 mm; $P < 0.001$). An increase in the following parameters was noted in concussion vs age-matched controls: dilation release amplitude (0.54 ± 0.96 vs 0.36 ± 0.05 mm; $P < 0.001$) and latency (0.25 ± 0.05 vs 0.21 ± 0.02 s; $P < 0.001$). Subjects with concussion with photosensitivity exhibited increased dilation release amplitudes ($P < 0.001$).

Conclusions: Individuals with sport concussion had impairment in pupillary constriction velocities, latency, and diameter in more than 1 year after concussion. The increase in dilation release amplitude among subjects with concussion might serve as a biomarker in diagnosing the underlying symptom of photosensitivity. The iPhone-based pupillometry could serve as a convenient and diagnostic tool in diagnosing these symptoms.

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A concussion happens when the brain is subjected to a direct or indirect force that alters brain function.[1] The estimated number of visits by people aged 19 or younger for sports-related and recreation-related concussions to hospital emergency departments increased 62% between 2001 and 2009, from 153,375 to 248,418, according to the Centers for Disease Control and Prevention.[2] In a four-triad model of mild traumatic brain injury (mTBI) proposed by Ciuffreda et al,[3] various oculomotor and nonoculomotor visual consequences of mTBI were discussed. Among nonoculomotor visual issues, photosensitivity is one of the commonest complaints observed after a concussion. A general definition of photosensitivity is a visual symptom of mild to severe visual discomfort experienced by a person under normal lighting conditions (i.e., light levels that normally do not trigger visual discomfort in others).[4] Unfortunately, there has not been many research that specifically address the phenomenon of photosensitivity in this diagnostic criterion. Photosensitivity and its prevalence (i.e., 10% in normals and 50% in concussion) have frequently been discussed in studies on concussion, but mainly as a general survey of the total visual sequelae and related components.[5,6] Because there is no known objective test or objective biomarker to conclusively demonstrate the presence of photosensitivity, the diagnosis of the condition might remain unclear. Therefore, it is of utmost importance to fill the knowledge gap by understanding the possible objective abnormal pupillary responses to light in photosensitive subjects with concussion so that effectiveness of management options such as tinted glass could be known with due course.

Moreover, the pupillary light reflex (PLR) has been studied previously in humans and has been found to be impaired in many cases of mTBI for a variety of criteria.[7,8] Given that the PLR represents how the visual system interprets light and that photosensitive people interpret light differently from other people, there may be a difference in the PLR dynamics between those with and without photosensitivity. In a recent study, photosensitive subjects were found to have faster pupillary dilation velocity and a larger diameter, when assessed with an infrared objective pupillometry technique.[9] This study thus aimed at evaluating the pupillary dynamics using an iPhone-based pupillometry technique in individuals with concussion with photosensitivity and comparing them with aged-matched controls.

Ridley Collge of Optometry & Srimanta Sankaradeva University of Health Sciences, Assam, India.

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Address correspondence to Pritam Dutta, MOptom, Ridley College of Optometry, K Handique Path, Jorhat 785001, Assam, India; E-mail: pdutta029@gmail.com

METHODS

This prospective study was conducted among 50 athletes and 50 aged-matched healthy controls. The project has been approved by the institutional review board and ethics committee (CEH-3012-IEC, Chandraprabha Eye Hospital) and has also followed the guidelines proposed by the Declaration of Helsinki. The athletes were male football players who had a history of sport-related concussion with a duration of 1 year or more and persistent concussive symptoms and an adequate visual acuity of 20/30 or better. Uncorrected refractive error, pre-existing ocular and health issues, and under any ocular and systemic medications affecting pupillary dynamics were excluded. Because the athletes with concussion were male, age-matched controls considered were also male, with no history of concussion, visual acuity of 20/30 or better, and stable ocular and general health.

Instrumentation

The Reflex-Pro PLR analyzer application is a mobile-based automated pupillometer that provides medical practitioners with a quantitative technique to evaluate pupil size and reactivity. According to the FDA's 21 CFR 886.1700, Reflex is a medical device that is Class I 510 (k) exempt. With iOS infrastructure and a strong mobile or wireless connection, it collects a video of a subject's eye at 30 Hz, parses it into images, and then sends those images to a trained object detector to quickly identify an iris. The iPhone Reflex app generates a photic flash stimulus with a length of about 1 millisecond (ms) and 10 lumens set to 70% power (flash brightness), and records dynamic reactions at 30 Hz or 30 frames per second. The subjects were asked to position their eyes so that they filled the almond-shaped window at the start of the test (Fig. 1). When the test subject was prepared, the examiner tapped the spherical soft button at the bottom center of the display that has 0% in the center. The examination was performed in a clinical setting in a dim room illumination. All the recordings were taken on the same day with constant room illumination of 4 lux by the same examiner. The intensity of the light source was 50 μW for a duration of 5 seconds.

Post-Concussion Symptom Scale

The Post-Concussion Symptom Scale (PCSS) questionnaire was used to assess subjective symptom evaluation. It is a 22-item scale used to measure the severity of symptoms after a concussion. This scale was developed with the goal of assigning a numerical value to each symptom to objectively quantify the common subjective symptoms that an individual experiences after experiencing a concussion. The purpose of designing this scale was to supplement other techniques such as neuropsychological testing. Each symptom was scored on a scale of 0–6. The higher the scores, the more severe the symptoms.

Statistical Analysis

Analyses were conducted using IBM SPSS (Statistical Package for Social Sciences) version 20.0. Normality distributions of the data were analyzed using the Shapiro–Wilk test. Descriptive statistics of mean and SD were used for age calculation and median/interquartile range (IQR) was used for denoting PCSS scores. Comparison of pupillary parameters between subjects with concussion and controls were performed using the independent *t* test with 95% confidence interval (CI). *P*-values are calculated on a 2-tailed basis. Comparison of pupillary parameters between subjects with concussion with and without photosensitivity was performed using the Mann–Whitney test and was denoted using a box and whisker graph.

RESULTS

Fifty athletes with concussion and 50 age-matched controls were included in the study. The mean ± SD age of the subjects was 27 ± 4 years and 26 ± 5 years in the concussed and aged-matched control groups (P = 0.45), respectively. The mean ± SD duration of the concussion injury among athletes with concussion was 19 ± 5 months.

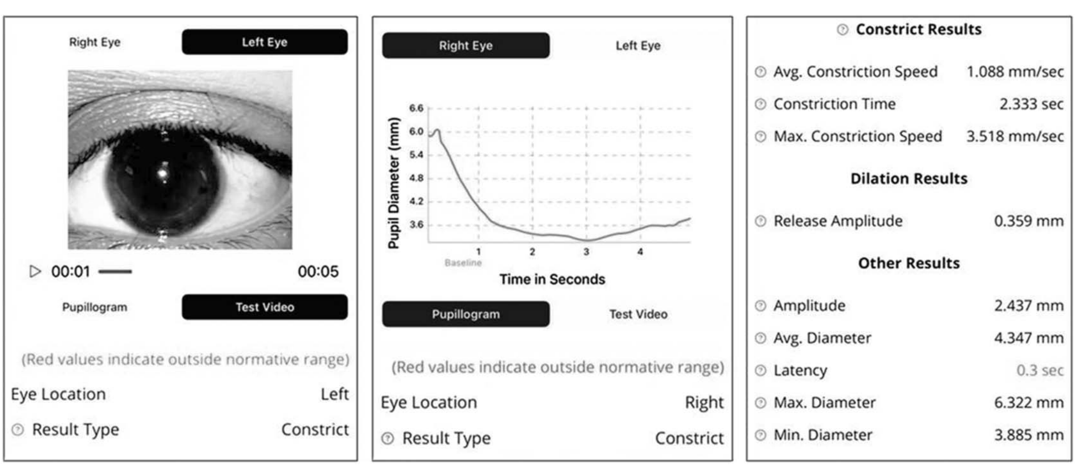

FIG. 1. Snapshot of the screen of Reflex-Pro PLR analyzer on focusing the eye and interpreting the results on display.

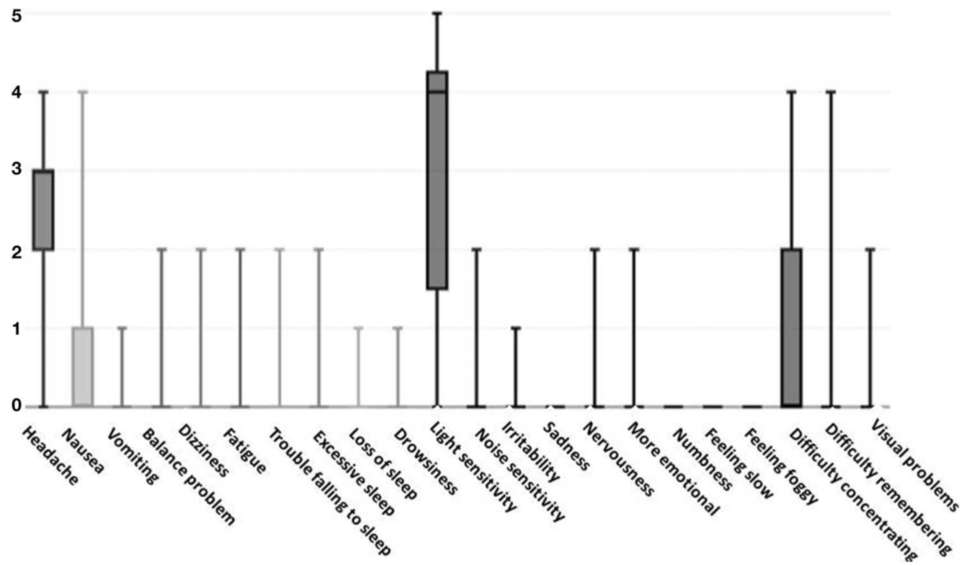

FIG. 2. Box and whisker plot showing the Post-Concussion Symptom Scale (PCSS) scores among subjects with concussion.

Post-Concussion Symptom Scale Response

The most common symptoms reported by the subjects with concussion were headache (n = 21) with a median (IQR) score of 3 (1) and light sensitivity (n = 29) with a median (IQR) score of 4 (3) (Fig. 2).

Pupillary Parameters Among Concussed vs Age-Matched Controls

There was a decrease in mean ± SD of following parameters in concussion vs age-matched controls: average constriction speed (1.10 ± 0.15 vs 1.78 ± 0.12 mm/s; $P < 0.001$), maximum constriction speed (2.05 ± 0.26 vs 3.84 ± 0.28 mm/s; $P < 0.001$), average diameter (3.64 ± 0.12 vs 0.36 ± 0.05 mm; $P < 0.001$), maximum diameter (4.75 ± 0.17 vs 5.23 ± 0.16 mm; $P < 0.001$), and minimum diameter (2.75 ± 0.17 vs 3.64 ± 0.11 mm; $P < 0.001$). An increase in the following parameters was noted in concussion vs age-matched controls: dilation release amplitude (0.54 ± 0.96 vs 0.36 ± 0.05 mm; $P < 0.001$) and latency (0.25 ± 0.05 vs 0.21 ± 0.02 s; $P < 0.001$) (Table 1). On further analysis between the pupillary parameters between subjects with concussion with and without photosensitivity, it was found that none but only pupillary dilation release amplitude was significantly different. Median (IQR) of dilation release amplitude among subjects with concussion with and without photosensitivity includes 0.33 (0.31–0.36) and 0.66 (0.55–0.70) (Z-score = 6.58; $P < 0.001$, Mann–Whitney U test) (Figs. 3, 4).

TABLE 1. Comparison of pupillary parameters between subjects with concussion and controls

Pupillary Parameters	Group	Mean ± SD	Range	Std. Error Difference	95% Confidence Interval of the Difference	P
Average constriction speed, mm/s	Controls	1.78 ± 0.12	1.43–2.00	0.03	0.59 (0.74)	<0.001
	Concussed	1.10 ± 0.15	0.75–1.45			
Maximum constriction speed, mm/s	Controls	3.84 ± 0.28	3.05–4.32	1.78	1.64 (1.92)	<0.001
	Concussed	2.05 ± 0.26	1.65–2.65			
Dilation release amplitude, mm	Controls	0.36 ± 0.05	0.29–0.52	−0.18	−0.22 (−0.14)	<0.001
	Concussed	0.54 ± 0.96	0.36–0.70			
Average diameter, mm	Controls	4.14 ± 0.18	3.68–4.56	−0.50	−0.58 (4.041)	<0.001
	Concussed	3.64 ± 0.12	3.33–3.67			
Latency, s	Controls	0.21 ± 0.02	0.17–0.26	−0.12	−0.16 (−0.11)	<0.001
	Concussed	0.25 ±0.05	0.24–0.33			
Maximum diameter, mm	Controls	5.23 ± 0.16	4.93–5.57	−0.48	−0.57 (−0.39)	<0.001
	Concussed	4.75 ± 0.17	4.45–5.12			
Minimum diameter, mm	Controls	3.64 ± 0.11	3.25–3.86	−0.91	−0.99 (−0.83)	<0.001
	Concussed	2.72 ± 0.17	2.45–3.02			

CONCLUSIONS

Pupillary examination tools have been introduced recently and are extensively used in ocular and neurologic research.[8–11] This article emphasizes the role of a novel approach to pupillary testing and its effects after a concussion. Constriction latency was found be significantly delayed in concussion. This is in accordance with the observations noted by Truong and Ciuffreda[12] which showed 5 of 6 test conditions to be significantly reduced in subjects with mTBI, consistently for less intense stimuli. However, the inverse relation between stimulus intensity and latency is also well established, meaning when the stimulus is less intense, the latency is greater, and vice versa.[13] The current investigation also used a less intense stimulus, under which subjects with concussion exhibited a statistically significant reduction in the latency period. This could be the result of a response saturation effect that would inadvertently obscure any transient, subtle response variations.

This study also found pupil diameters (maximum, minimum, and average) to be reduced in subjects with concussion. Because the pupillary diameter is a function of the Edinger–Westphal nucleus (EWN) level of midbrain supranuclear inhibition, the existence of a smaller pupillary diameter in concussion under the current test settings could be possibly due to a reduction in EWN inhibition.[14,15] A similar study by Master et al has also reported subjects with sports concussion exhibiting larger pupillary diameters at their acute stage of injury, that is, ≤7 days.[16] These discrepancies in the pupillary diameters after a concussion could be possibly due to variations in the duration in which the subjects were recruited after a concussion. The present study recruited subjects after a duration of 1 year of concussion to avoid alteration of outcomes from the natural recovery process; however, in the later investigation by Master et al, subjects with concussion were recruited in their acute stage, that is, ≤ 7 days postinjury.[16]

Constriction velocities are considered to be the most reliable metrics for identifying parasympathetic dysfunction.[18] The maximum constriction velocity reflects the rapid neurologic reaction element to a stimulus (open-loop), whereas the average constriction velocity reflects the remaining slower response component averaged over time under visual feedback control (closed-loop).[17] The reduction in these velocities among the subjects with concussion observed in this study reflects a poor rapid and slow neural

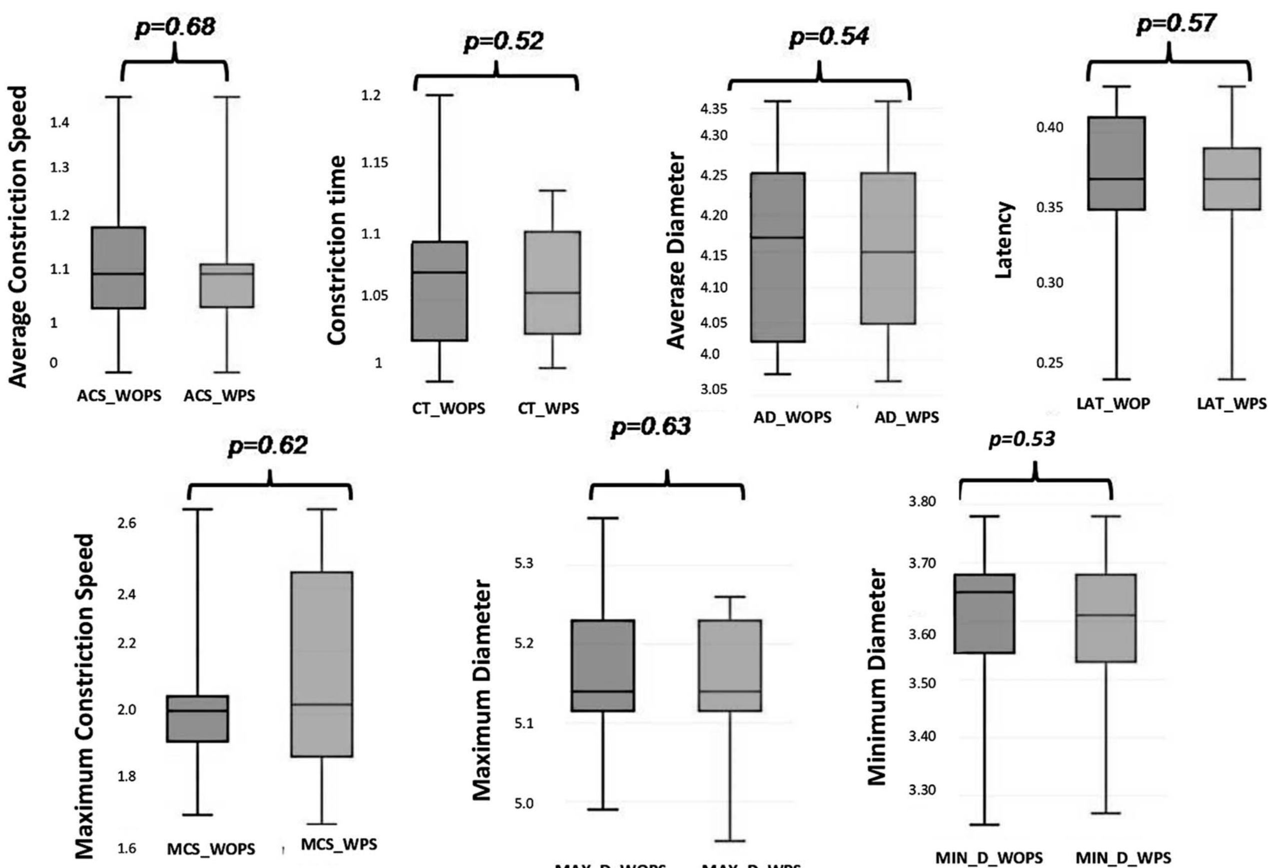

FIG. 3. Box and whisker plot showing the median (IQR) of pupillary parameters between subjects with concussion with (WPS) and without photosensitivity (WOPS).

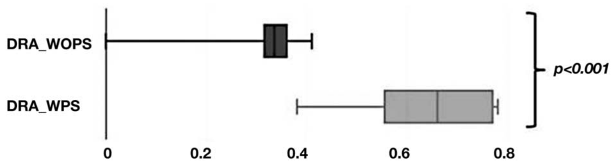

FIG. 4. Box and whisker plot showing the dilation release amplitude among subjects with concussion with (DRA with PS) and without photosensitivity (DRA without PS).

reflex response. One study by Master et al found that the constriction velocities are significantly increased in sports concussion at their acute phase of presentation,[16] whereas another study by Thiagarajan and Ciuffreda[8] found subjects with concussion after 1 year of their initial head injury exhibiting a statistically significant reduction in the constriction velocities when compared to subjects with no history of concussion. These changes in the constriction velocities observed in the present study and in the earlier investigations serve as a potential diagnostic factor for concussion. It could be hypothesized that in the acute phase of concussion, the pupillary constriction velocities do not get reduced immediately. Because there is a diffuse shearing and sheathing of the neurons in concussion, the neural signal processing could be reduced with time, thus reducing the amount of information processing for pupillary constriction and hampering the constriction velocities with time.[19]

The poor constriction velocities among the subjects with concussion also led to faster redilation. Moreover, on further subanalysis between subjects with concussion with and without photosensitivity, it was found that none but only the dilation release amplitude was significantly increased in those with photosensitivity. However, this rapid redilation in subjects with concussion allows more light into the eye, as well as subsequent associated neural pathways, resulting in photosensitivity. In contrast to this finding, Troung et al in their investigation portrayed that when the baseline light level sensor is affected, there occurs a reduction in the sustained constriction response (greater redilation).[9] The intrinsically photosensitive retinal ganglion cells (ipRGCs) would most likely be involved if the baseline light sensor failed.[20] This could be presumed that reduction in the robustness of the ipRGC system occurs because of concussion, as evidenced by a poor constriction response, resulting in photosensitivity. Because the control groups in the analysis were not photosensitive, it could serve as a potential source of limitation of the present study. However, the present findings could serve as a baseline for further investigations with a larger sample to compare the pupillary dynamics change in photosensitive subjects with and without concussion for better understanding. Thus, it could be concluded that pupillary constriction velocities and latencies are impaired in the case of concussion. The increase in dilation release amplitude among subjects with concussion might serve as a biomarker in diagnosing the underlying symptom of photosensitivity. This iPhone-based pupillometry may be a potential diagnostic tool in long-term photophobia after concussion.

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